Database Schema Design

Digital Wardrobe & Outfit Recommendation System (MySQL / Relational Schema)

DBMS VIVA REFERENCE

Table Name	Attributes & Constraints	Primary Key	Foreign Keys & Cascades	Description
USER	**id** (INT), **email** (VARCHAR 255, UNIQUE), **name** (VARCHAR 120), **password_hash** (VARCHAR 255), **created_at**, **updated_at** (DATETIME)	`id`	—	Stores registered user credentials, profile display name, and authentication digests.
CATEGORY	**id** (INT), **name** (VARCHAR 50, UNIQUE), **slug** (VARCHAR 50, UNIQUE), **wear_limit** (INT), **sort_order** (INT), **created_at** (DATETIME)	`id`	—	Classifies clothing types (Topwear, Bottomwear, etc.) and houses wear limits triggering laundry.
BRAND	**id** (INT), **name** (VARCHAR 80, UNIQUE), **created_at** (DATETIME)	`id`	—	Lookup catalog for garment brands and manufacturers to eliminate text redundancy.
SEASON	**id** (INT), **name** (VARCHAR 30, UNIQUE), **slug** (VARCHAR 30, UNIQUE), **sort_order** (INT), **created_at** (DATETIME)	`id`	—	Seasonal domain catalog (Summer, Winter, Rainy, Spring) driving seasonal colour filtering.

CLOTHINGITEM	**id** (INT), **user_id**, **category_id**, **brand_id**, **season_id**, **name** (VARCHAR 120), **color** (VARCHAR 30), **formality_score** (INT 1-5), **purchase_price** (DECIMAL 10,2), **status** (VARCHAR 20, CLEAN/LAUNDRY), **wear_count**, **wear_count_since_wash**, **last_washed_at**, **sent_to_laundry_at**, **image_url**, **created_at**, **updated_at**	`id`	`user_id → USER(id) [CASCADE]`
			`category_id → CATEGORY(id) [RESTRICT]`
			`brand_id → BRAND(id) [SET NULL]`
			`season_id → SEASON(id) [SET NULL]`
				Core inventory entity storing clothing attributes, laundry state, wear metrics, price, and photo pointers.

OUTFIT	**id** (INT), **user_id** (INT), **name** (VARCHAR 120), **occasion** (ENUM: CASUAL, SMART_CASUAL, WORK, FORMAL), **notes** (TEXT), **created_at**, **updated_at** (DATETIME)	`id`	`user_id → USER(id) [CASCADE]`	Stores saved or generated outfits along with target occasion rules and optional styling notes.
OUTFITITEM	**id** (INT), **outfit_id** (INT), **item_id** (INT), **position** (INT: 1=Base, 2=Layer, 3=Footwear) • UNIQUE(outfit_id, item_id)	`id`	`outfit_id → OUTFIT(id) [CASCADE]` `item_id → CLOTHINGITEM(id) [CASCADE]`	Bridge / Junction table resolving M:N relationship between Outfits and Clothing Items.

WEARLOG	**id** (INT), **user_id** (INT), **item_id** (INT), **outfit_id** (INT, NULLABLE), **date_worn** (DATE), **weather_temp** (INT), **notes** (TEXT), **created_at** (DATETIME)	`id`	`user_id → USER(id) [CASCADE]`
			`item_id → CLOTHINGITEM(id) [CASCADE]`
			`outfit_id → OUTFIT(id) [SET NULL]`
				Time-series log of wears powering wear counters, Cost-Per-Wear (CPW), and laundry auto-flagging.

COLORCOMPATIBILITY	**id** (INT), **color1** (VARCHAR 30), **color2** (VARCHAR 30), **score** (INT, 0-10) • UNIQUE(color1, color2) where color1 ≤ color2	`id`	—	Pairwise colour harmony lookup matrix queried using LEAST() and GREATEST() in outfit generator.

Digital Wardrobe System • Relational Schema & Constraint Documentation

3NF / BCNF Normalization • Foreign Keys with Integrity Cascades • MySQL Schema